

Quantifying the effects of temporal autocorrelation on climatological regression models using geostatistical techniques

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Abstract: When time-dependent data are used in regression models, temporal autocorrelation violates ordinary least squares assumptions and impedes their proper testing and interpretation. The problem of temporal autocorrelation is exacerbated by the uneven temporal spacing inherent in many data sets. Using simple linear regression models of stomatal conductance as examples, we compare the effectiveness of two methods for removing temporal autocorrelation from regression models (first-differencing and Cochrane–Orcutt) and we introduce the geostatistical technique of semivariograms as a method for quantifying temporal autocorrelation in uneven time series. The Cochrane–Orcutt method proved more effective than first-differencing at removing autocorrelation and produced regression models without changing the significance of the independent variables. Semivariograms were used to quantify the time dependence of the unevenly spaced stomatal conductance time series. This technique revealed the dominant autocorrelation at the minimum time lag (0.5 h) and the 24-h periodicity caused by the climatological variables used in the model. We conclude that geostatistical techniques provide a robust method for quantifying temporal structure and periodicity in unevenly spaced time series.

Résumé : L'utilisation de données dépendantes dans le temps dans les modèles de régression invalide les tests et leur interprétation parce que l'autocorrélation temporelle présente dans ces données viole les prémisses des moindres carrés ordinaires. Le problème de l'autocorrélation temporelle est exacerbé par les intervalles de temps inégaux inhérents à plusieurs séries de données. En prenant comme exemples des modèles simples de régressions linéaires de conductance stomatique, nous comparons l'efficacité de deux méthodes (différence de premier ordre et Cochrane–Orcutt) pour éliminer l'autocorrélation temporelle des modèles de régression, et nous introduisons la technique géostatistique des semi-variogrammes comme méthode pour quantifier l'autocorrélation dans des séries temporelles inégales. La méthode Cochrane–Orcutt s'est avérée plus efficace que la méthode de différence de premier ordre pour éliminer l'autocorrélation temporelle, et a produit des modèles de régression sans modifier le degré de signification des variables indépendantes. Nous avons aussi utilisé les semi-variogrammes pour quantifier la dépendance temporelle dans la série temporelle inégale de mesures de conductance stomatique. Cette technique a mis à jour la dominance de l'autocorrélation au décalage minimum (0,5 h) et la périodicité de 24 h engendrée par les variables climatologiques utilisées dans le modèle. Nous en concluons que les techniques géostatistiques sont des méthodes robustes pour quantifier la structure temporelle et la périodicité dans les séries temporelles inégales.

[Traduit par la Rédaction]

Introduction

Solar radiation, vapour pressure deficit, and air temperature are typical climatological variables that exhibit strong temporal autocorrelation in diurnal cycles. When variables like these are used in regression models it is important to account for temporal autocorrelation, because even though autocorrelation does not affect the unbiasedness or consistency of the ordinary least squares (OLS) regression estimators, it does affect their efficiency. Time-series models used for the purpose of prediction commonly retain

temporal autocorrelation to improve predictive ability. For example, autoregressive moving average (ARMA) and autoregressive integrated moving average (ARIMA) methods (Diggle 1990) are popular in predictive modeling. However, if the goal of regression modeling is for hypothesis testing or to determine the structure of underlying deterministic processes, then autocorrelation must be quantified and removed because it can mask these underlying deterministic relationships. Detecting and removing autocorrelation from time series is difficult when the data are unevenly spaced. This is a common but often unavoidable situation that happens when some observations are missing, outliers are removed, sampling is irregular, or the phenomenon occurs at irregular times.

The specific aims of this study are (i) to compare the effectiveness of methods for removing temporal autocorrelation from regression models and (ii) to introduce

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semivariograms as a method for quantifying temporal autocorrelation in uneven time series. Semivariograms, unlike conventional spectral methods, can be calculated from unevenly spaced time-series data. Furthermore, they do not require stationarity of the underlying stochastic processes (Herzfeld 1990) and have been shown to reveal relatively more of the signal in situations of heavy noise and extreme outliers (Mayer 1993).

Two regression models of stomatal conductance are developed for trembling aspen (*Populus tremuloides* Michx.) and white birch (*Betula papyrifera* Marsh.) from data collected at irregular time intervals during a single growing season. The models are first tested for autocorrelation and its magnitude is quantified. We then compare first-differencing and Cochrane–Orcutt methods for removing autocorrelation from the regression models. Finally, we demonstrate that the dominant periodicity of the temporal autocorrelation in unevenly sampled stomatal conductance can be estimated using semivariograms.

Temporal autocorrelation in regression models

A simple linear regression model with time-series data can be expressed as

$$[1] \quad Y_t = \beta_0 + \beta_1 X_t + U_t$$

where Y_t and X_t are observations for period t and U_t is an error (or residual) term. One of the assumptions of OLS estimation is that the expected value of the error term covariances is zero (i.e., the residual terms are assumed not correlated). If the residuals do not fulfill this assumption of mutual independence, they are said to be autocorrelated (Dutta 1975). In this case, the regression estimates are unbiased, but inefficient (Bennett 1979). Neter et al. (1983) observed that if the error terms in the regression model are positively autocorrelated then the use of OLS procedures has several further important consequences: the mean squared error may substantially underestimate the variance of the error terms, the standard deviation of the estimated regression coefficient calculated according to OLS may underestimate the true standard deviation, and the confidence intervals and tests using t and F distributions are biased. The last of these is perhaps the most important, because previous studies have used these statistics (t and F) to determine the form and strength of regression models (e.g., Krishnan and Kushwaha 1973; Saylan and Bernhofer 1993).

One approach to evaluate the nature and extent of temporal autocorrelation in regression models is to quantify the relationship between successive residuals. For example, autocorrelation among residuals of climatological regression models has been studied in the context of economic forecasting and decision-making problems that require climate as an input (Katz 1993). If it is assumed that the residuals are linearly dependent according to the first-order Markov sequence

$$[2] \quad U_t = \rho U_{t-1} + V_t$$

then the residual term at time t (U_t) consists of two parts. One is systematically dependent on U_{t-1} and explained by the coefficient ρ , and the other, V_t (an arbitrary disturbance),

is truly random and fulfills the basic assumptions of OLS, i.e., zero mean, mutual independence, and constant and finite variance (Dutta 1975). The estimate of the first-order serial correlation coefficient ($\hat{\rho}$) is given by

$$[3] \quad \hat{\rho} = \frac{\sum \hat{U}_t \hat{U}_{t-1}}{\sum \hat{U}_{t-1}^2}$$

When autocorrelation is present, the initial remedial step is to check for missing independent variables and, failing this, to transform the variables. If the model has been mis-specified such that one or more key variables have been omitted from the model, the error terms in the regression model will tend to be positively autocorrelated, since the error terms include effects of the missing variables (Neter et al. 1983).

Methods for testing for temporal autocorrelation and quantifying its structure are described elsewhere (e.g., Diggle 1990; Hamilton 1994; Hipel and McLeod 1994). The Durbin–Watson statistic is a measure of the first-order autocorrelation in residuals, and the autocorrelation function (ACF) and partial autocorrelation function (PACF(k)) are useful tools for quantifying the correlation between time-series observations k units apart after controlling for the correlation at intermediate lags (McDowall et al. 1980).

Correcting for autocorrelation in regression models

First-differencing method

The special case of ρ equal to 1 leads to a commonly used estimation procedure known as first-differencing (Dutta 1975). This allows a transformation of the independent and dependent variables at every time t according to

$$[4] \quad X_t^* = X_t - X_{t-1}$$

and

$$[5] \quad Y_t^* = Y_t - Y_{t-1}$$

where X^* and Y^* are the transformed variables. The transformed variables are used in the new regression equation as

$$[6] \quad Y_t^* = \beta_0^* + \beta_1^* X_t^* + U_t^*$$

with

$$[7] \quad U_t^* = U_t - U_{t-1} = V_t$$

and

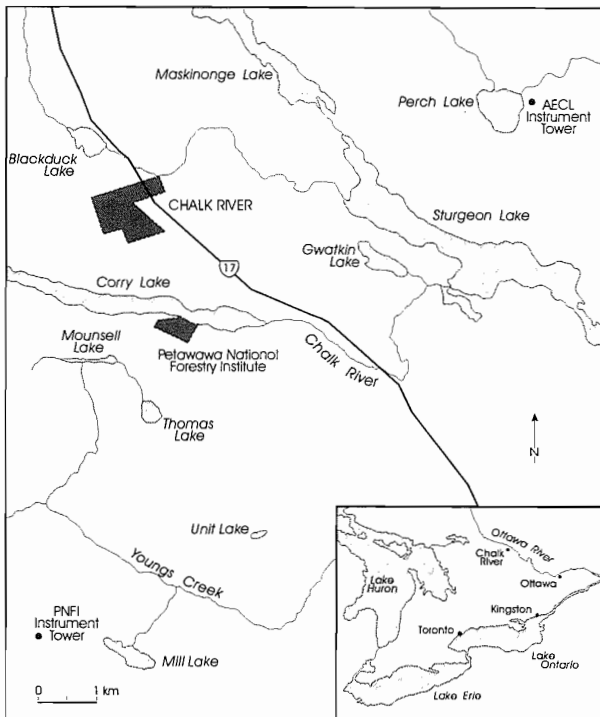
$$[8] \quad \beta_0^* = \beta_0 (1 - \rho) = 0$$

which shows how first-differencing eliminates the need for a constant term in the transformed equation. The diagnostic statistics can be reapplied to determine whether further transformations are necessary.

Cochrane–Orcutt method

The simplifying assumption that $\rho = 1$ does not necessarily produce the best transformations of dependent and independent variables. The Cochrane–Orcutt method proposes that the Markov estimator be based on residuals from the first regression. Equation 3 is used to estimate ρ and to transform the original observations of X and Y following

$$[9] \quad X_t^* = X_t - \hat{\rho} X_{t-1}$$

Fig. 1. Location of the AECL forest.

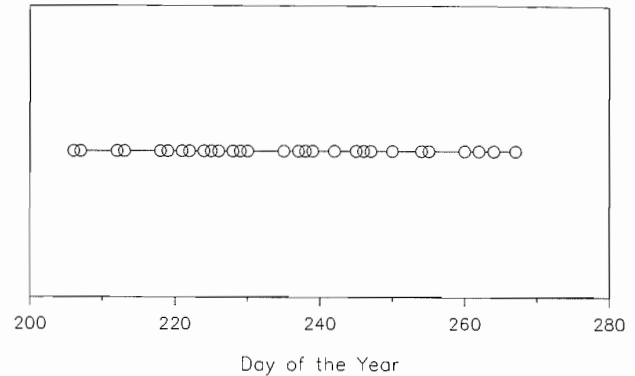
and

$$[10] \quad Y_t^* = Y_t - \hat{\rho}Y_{t-1}$$

The regression can then be repeated (as with first-differencing) following eq. 6, using diagnostic statistics to determine the relative success of the procedure.

Quantifying temporal autocorrelation in uneven time series

It is beyond the scope of this paper to review the numerous techniques based on the Fourier transform and spectrum estimation to analyze time series (see Diggle 1990; Hamilton 1994; Hipel and McLeod 1994). However, it is important to note that all of these methods typically require equally spaced samples. Diggle (1990) and Herzfeld (1990) have demonstrated that geostatistical methods can be used to analyze unevenly spaced time series, but applications are rare. Geostatistical methods were developed originally for spatial data (Cressie 1991; Dowd 1984; Haining 1990), but are now used in soil science (McBratney and Webster 1986; Robertson and Gross 1994), ecology (Freire et al. 1992; Jackson and Caldwell 1993a, 1993b; Phillips 1985; Rossi et al. 1992; van Es 1993), coastal geomorphology (Young and Harvey 1996), forestry (Biondi et al. 1994), hydrology (Li and Lake 1994), and most recently in cyclic stratigraphy (Mayer 1993). We calculate semivariograms for the observed unevenly spaced time series of aspen and birch stomatal conductances to quantify their time-dependent structure.

Fig. 2. The uneven nature of the time series. Circles indicate days on which stomatal conductance data were collected in 1989.

Geostatistics is based on the theory of regionalized variables, which recognizes that values nearby a measured location are more alike than those measured farther apart. The semivariogram function, $\gamma(k)$, is used to formalize the spatial structure of a random variable, Z , by quantifying the variance between pairs of observations as a function of their separation distance and direction, or time lag, k (van Es 1993):

$$[11] \quad \gamma(k) = 0.5 \text{ var } (Z_{i+k} - Z_i)$$

In other words, the $\gamma(k)$ relationship explicitly recognizes that the set of random variables, Y_p , is characterized by a structure that can be quantified in an estimate of the variance (or covariance) defined as

$$[12] \quad \text{var}(Z_{i+k} - Z_i) = E(Z_{i+k} - Z_i)^2 = 2\gamma(k)$$

for all observations at time lag k apart (van Es 1993). The standard semivariogram is then estimated as

$$[13] \quad \hat{\gamma}(k) = \frac{1}{2N(k)} \sum_{i=1}^{N(k)} (Z_{i+k} - Z_i)^2$$

where $N(k)$ is the number of pairs of observations separated by time lag k . Several estimators of the variance have been proposed (Cressie 1991; Haining 1990; Li and Lake 1994), but the median absolute difference estimator was chosen for this study because it is resistant to outliers (Haining 1990). This estimator is calculated as

$$[14] \quad 2\gamma(k) = 2.198[\text{median}|y(t_i) - y(t_{i+k})|]^2$$

where 2.198 is a correction factor (Haining 1990) and medians are taken over all pairs separated in time by a difference of k . When semivariance values are plotted against corresponding k values, the resultant semivariogram has a characteristic structure that represents the degree of similarity or dissimilarity between samples.

Analysis of semivariance, $\gamma(k)$, is arguably one of the most versatile and unbiased means of examining autocorrelation in environmental data (z -space domain), but until recently it has not been used on time series (spectral domain). The main statistical advantages of using semivariance over other measures of autocorrelation are that

Table 1. Regression results from the original model, the first-differencing transformation model, and the Cochrane–Orcutt transformation model for aspen and birch.

Model		Coefficient	SE	<i>t</i>	<i>P</i> (2-tail)	<i>n</i>	<i>F</i> ratio	<i>d</i>	FOA
Aspen species									
Original	Constant	−1.477	0.076	−19.0	0.000	322	160.4	1.185	0.399
	<i>K</i>	0.001	0.000	14.0	0.000				
	TA	0.044	0.005	9.5	0.000				
	VPD	−0.674	0.039	−17.0	0.000				
First-differencing	Constant	0	na	na	na	219	4.11	2.259	−0.137
	<i>K</i>	0.0003	0.000	2.67	0.008				
	TA	−0.039	0.034	−1.15	0.253				
	VPD	0.272	0.177	1.54	0.126				
Cochrane–Orcutt	Constant	−0.722	0.056	−13.0	0.000	219	58.5	2.051	−0.026
	<i>K</i>	0.001	0.000	8.4	0.000				
	TA	0.059	0.007	8.3	0.000				
	VPD	−0.745	0.069	−11.0	0.000				
Birch species									
Original	Constant	−1.509	0.084	−18.0	0.000	322	183.6	0.778	0.604
	<i>K</i>	0.0008	0.000	11.0	0.000				
	TA	0.065	0.005	13.0	0.000				
	VPD	−0.905	0.043	−21.0	0.000				
First-differencing	Constant	0	na	na	na	219	5.49	2.389	−0.195
	<i>K</i>	0.0002	0.000	2.82	0.005				
	TA	0.039	0.032	1.26	0.210				
	VPD	−0.479	0.165	−2.91	0.004				
Cochrane–Orcutt	Constant	−0.335	0.053	−6.3	0.000	219	22.7	2.160	−0.081
	<i>K</i>	0.0004	0.000	4.4	0.000				
	TA	0.081	0.015	5.6	0.000				
	VPD	−0.921	0.121	−7.6	0.000				

Note: All *F* ratios are significant at the 0.05 level. *d*, the Durbin–Watson test statistic; FOA, the first-order autocorrelation.

it can be used on unevenly spaced data and it does not require second-order stationarity in kriging estimations (Robertson and Gross 1994). Spectral techniques are possible in these situations only after a high degree of pre-processing of the original data (interpolation and resampling at a regular interval, filtering, inversion), which introduces smoothing and rounding errors (Mayer 1993).

A climatological model of stomatal conductance

The study site

To demonstrate the above statistical techniques, we use climatological data derived from a long-term research project in forest microclimatology conducted by Dr. J.H. McCaughey from Queen's University on the grounds of the Petawawa National Forestry Institute and Atomic Energy of Canada (AECL), both near Chalk River, Ontario (46°02'N, 77°20'W). The data were collected in a mixed-species deciduous forest at AECL (Fig. 1) during the summer of 1989. Trembling aspen and white birch are the

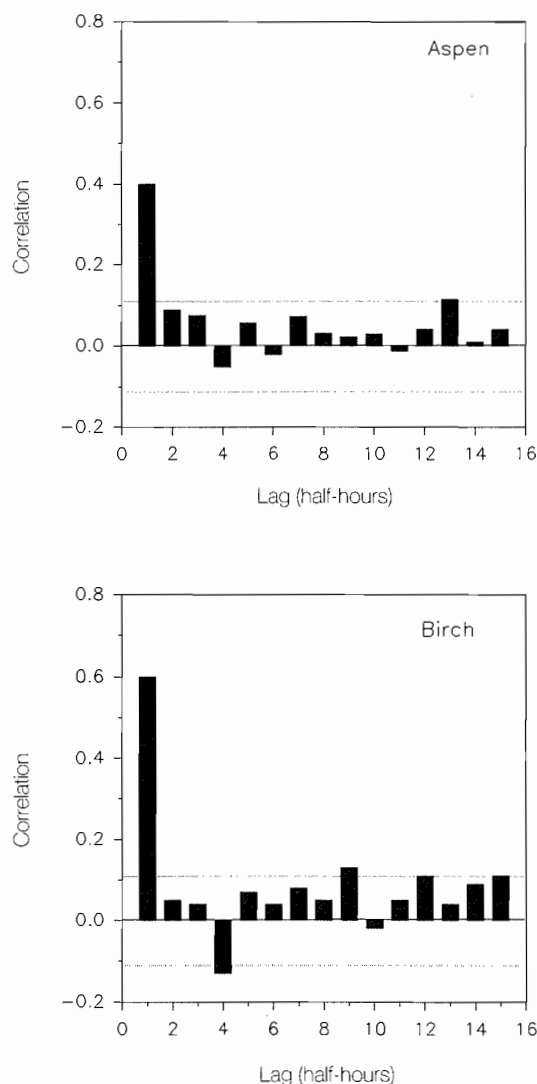
dominant overstorey species in this stand, composing more than 80% of the total stems.

Data collection and analysis

Stomatal conductance data of the dominant species were collected using a steady-state porometer (model 1600; LI-COR Inc.). A scaffold tower provided access to the foliage in the overstorey of two trees: one birch and one aspen. Ten leaves from each tree were sampled at half-hour intervals from a common layer (19–21 m) following the sampling strategy of Iacobelli and McCaughey (1993). These readings were averaged to obtain a single value for each species at half-hour intervals. The sampling was irregular with respect to time (Fig. 2), because porometry could not be conducted when the canopy was wet nor when concurrent experiments required maintenance.

Climatological data from above-canopy instrumentation mounted on the scaffold tower were recorded and averaged at 30-min intervals using a datalogger (CR7X, Campbell Sci.). Incoming solar radiation was measured by an Eppley

Fig. 3. PACF for aspen and birch residuals. Dotted lines indicate ± 2 standard errors.



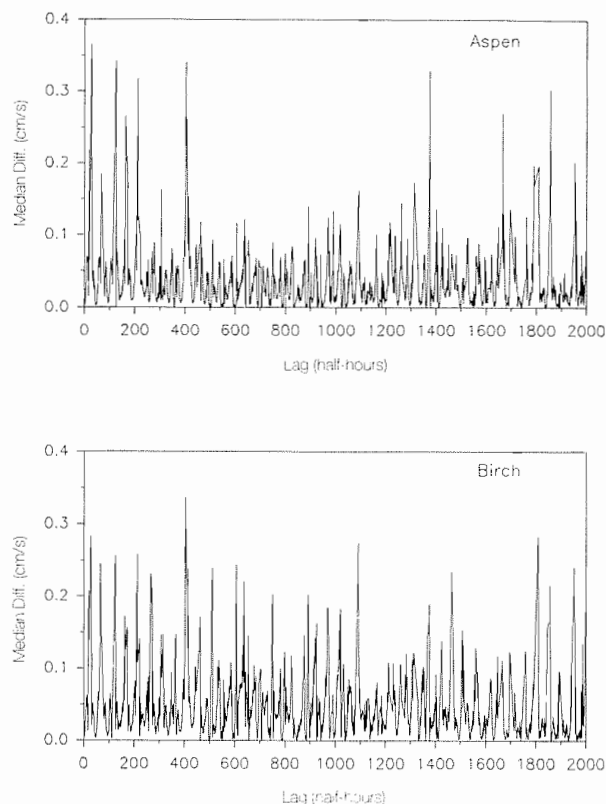
pyranometer mounted above the canopy. Air temperature and vapour pressure deficit were measured with aspirated psychrometers mounted at fixed levels within and above the canopy.

The linear regression models of stomatal conductance (g_s) have the general form

$$[15] \quad \ln(g_s)_t = \beta_0 + \beta_1 K_t + \beta_2 TA_t + \beta_3 VPD_t + U_t$$

where K_t is the incoming solar radiation at time t ($\text{W}\cdot\text{m}^{-2}$), TA_t is air temperature ($^{\circ}\text{C}$), VPD_t is vapour pressure deficit (mb), and U_t is a residual term. These independent variables are known to be the most significant climatological drivers of stomatal response (Iacobelli and McCaughey 1993; Stewart 1988). Other independent variables (such as a surrogate for soil moisture content) or different functional forms such as the phenomenological model first proposed by Jarvis (1976) could be used to produce a superior predictive model, but our

Fig. 4. Semivariograms of the median absolute difference conductance for aspen and birch for 2000 lags.



main interest here is to examine the effects of temporal autocorrelation on the regression models.

All statistical analyses (regressions and diagnostics) and curve fitting were performed using commercially available statistical packages. Median absolute difference semivariance was calculated for stomatal conductance based on eq. 14 using a C program written by one of us (L.E.H.).

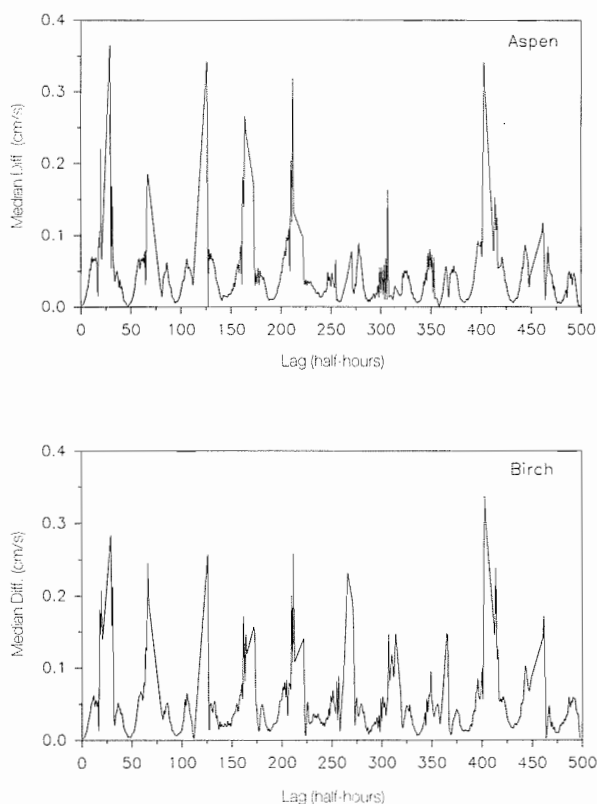
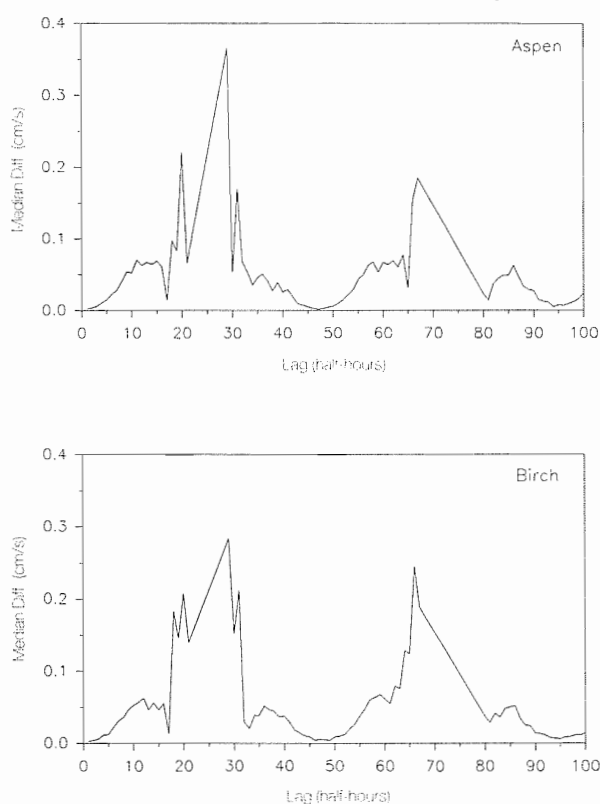
Results and discussion

Results for the regression models based on 1989 data are shown in Table 1. All climatological predictor variables are statistically significant, although these significance tests may be biased if autocorrelation is present. Durbin-Watson tests indicate that stomatal conductance for both species is autocorrelated as the observed d values lie below the lower limiting value, $d_L = 1.48$ ($P = 0.01$). The hypothesis of no positive serial correlation is thus rejected. The first-order autocorrelation is calculated as 0.399 for the aspen and 0.604 for the birch.

The PACF plot (Fig. 3) illustrates that autocorrelation is present among aspen and birch residuals. Significant autocorrelation occurs at the first time lag ($k = 1$) of 30 min (the minimum sampling time period).

The first-differencing method

The first-differencing method was used to remove autocorrelation from the regression models. Setting $\rho = 1$, the

Fig. 5. As in Fig. 4 but for the first 500 lags only.**Fig. 6.** As in Fig. 4 but for the first 100 lags only.

data were transformed according to eqs. 4 and 5, and a new regression equation was produced following eq. 6. The first-order autocorrelation was calculated as -0.161 and -0.213 for the aspen and birch regressions, respectively (Table 1). The hypothesis of no significant serial autocorrelation cannot be rejected for either species, because $4-d$ lies above the upper limiting value, $d_U = 1.60$ ($N = 219$; $P = 0.01$), for aspen and the test is inconclusive for birch. However, this technique changed the form of the models as TA is no longer significant for either model, and VPD is not significant for the aspen model.

The Cochrane–Orcutt method

The Cochrane–Orcutt method was used as an alternative to the first-differencing technique to eliminate autocorrelation from the regression models. Using the data sets with eq. 3 produced values for ρ of 0.518 and 0.782 for aspen and birch, respectively, after the second iteration (i.e., using the residuals from the original OLS iteration). The dependent and independent variables were then transformed according to eqs. 9 and 10 and modelled using eq. 6. After applying the Cochrane–Orcutt method, neither regression model shows significant autocorrelation (Table 1). The first-order autocorrelation is small (-0.026 for aspen and -0.081 for birch) and Durbin–Watson ($4-d$) tests indicate no significant autocorrelation ($d_U = 1.60$; $P = 0.01$). The model form is the same as the original; however, the estimated standard errors for the regression coefficients are

larger than those derived from the original regression. Therefore, the standard deviations derived for the original equations may have been underestimates of the true standard deviations.

Semivariograms

Semivariograms for both species are shown in Fig. 4. The important features of these semivariograms are the periodic dips representing similarity in observed stomatal response. Semivariograms for the first 500 lags (approximately 10 days) are displayed in Fig. 5. Variance is minimal at 48 half-hour time lags, corresponding to a periodicity on the order of 24 h, or a diurnal cycle. The diurnal signal is weaker with increasing lag times, because many factors operating on different time scales affect the climatological drivers, and thus the stomatal response. Although climatic conditions may remain constant from one day to the next, similar conditions will seldom persist beyond 3 or 4 days. Furthermore, uneven sampling reduces the number of data points at some time lags and weakens the semivariogram signal.

Semivariograms for the first 100 lags (Fig. 6) suggest a strong diurnal cycle and secondary cycle at approximately 17 half-hours (8.5 h) from the start of the primary cycle. The secondary dip in variance probably represents similar stomatal response within 1 day and the harmonic between successive days, because a time difference of 8.5 h represents those measurements taken in the morning and in the late afternoon. There are at least two possible explanations for

the similarity in stomatal response at these two times: (i) increasing stomatal response in the morning caused by the increase in solar radiation and (ii) a corresponding decrease in response in the afternoon due to the increase in vapour pressure deficit and the need to conserve water. In other words, this dip in variance may represent similar stomatal response to two different processes. Similar harmonic patterns in stomatal conductance from these same processes should also occur at a time lag of 34 half-hours (17 h; i.e., observations taken in the late afternoon on one day and observations taken in the morning of the next day). However, many observations at time lags between 8.5 and 17 h (and multiples thereof) are unavailable because porometry was not measured after sunset, when stomatal response is negligible.

A least-squares curve-fitting procedure was used to quantify the dominant time lag in stomatal conductance. All lags with fewer than five observations were first removed and a periodic function of the form

$$[16] \quad \gamma(k) = a \left(1 - \cos \frac{2\pi k}{b} \right)$$

(McBratney and Webster 1986) was fit to the first 100 time lags (approximately 2 days) using an iterative procedure. The constant b , representing the period of the function, was assigned an initial value of 48 half-hours. The iterative procedure converged at $a = 0.049$ and $b = 46.9$ ($r^2 = 0.866$) for aspen and $a = 0.051$ and $b = 46.6$ ($r^2 = 0.802$) for birch, confirming that the dominant periodicity of stomatal conductance is on the order of 1 day.

Conclusions and implications

We have used different techniques to detect, quantify, and remove temporal autocorrelation from multiple linear regression models of stomatal conductance. The Cochrane–Orcutt method is more effective at removing autocorrelation than the first-differencing technique, as indicated by the Durbin–Watson test and the first-order autocorrelation. Regression models with no temporal autocorrelation were produced without changing the significance of independent variables. The Cochrane–Orcutt technique involves a simple transformation of the data, so it is recommended when regression models are used to quantify deterministic relationships and test hypotheses.

Semivariograms, normally used in spatial analyses, were successfully used to quantify the cyclic nature of the unevenly spaced stomatal conductance time series. A strong diurnal cycle was detected, with minimal variance occurring at time lags of 48 half-hours (1 day). The unevenly sampled data could not be analyzed using spectral methods without many additional assumptions, but semivariograms confirmed that autocorrelation in stomatal response is dominant at the minimum time lag (1 half-hour) and exhibits a 24-h periodicity. This stomatal conductance periodicity is a function of the climatological variables, as demonstrated in the regression models. Similar climatological conditions and stomatal response are expected to persist for 2 to 3 days. In other words, the observed stomatal response at a given time on one day should be very similar to the observed stomatal response at the same time in successive

days. This time dependency of minimal variance at a lag of 48 half-hours is visible in the semivariograms and was quantified as on the order of 47 half-hours (or approximately 1 day) by fitting a periodic cosine function to the semi-variance values.

Temporal autocorrelation is potentially problematic for regression modeling when using time-dependent climatological variables. However, several parametric statistical techniques are available to test for the presence of temporal autocorrelation, quantitatively describe its structure, and remove its effects from the regression model. When the time series data are uneven, geostatistical techniques are recommended as robust and versatile tools to quantify the temporal structure and estimate the periodicity of underlying cycles. Unlike spectral techniques, bad data or outliers can be omitted to test their relative effect on the structure of temporal autocorrelation revealed in semivariograms.

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